Problem Set 2

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Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in `R`).

The null hypothesis is that the variables of class and solicitation of bribe are statistically independent. The alternative hypothesis is that the variables of class and solicitation of bribe are statistically dependent.

```r
# Create a contingency table with the data
cont_table <- matrix(c(14, 6, 7, 7, 7, 1), nrow = 2, ncol = 3, byrow =
    TRUE)
```

[1]Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multimethod Study of Bribery and Discrimination in Latin America. *Latin American Research Review.* 45 (1): 76-97.

```r
rownames(cont_table) <- c("Upper Class", "Lower Class")
colnames(cont_table) <- c("Not Stopped", "Bribe Requested", "Stopped/Given Warning")

# Chi Square by hand
chi_square_by_hand <- function(x) {
  if (!is.matrix(x)) {
    stop()
  } else {

    # Expected Frequencies
    fe_1 <- (sum(x[,1])/sum(x))*sum(x[1,])
    fe_2 <- (sum(x[,2])/sum(x))*sum(x[1,])
    fe_3 <- (sum(x[,3])/sum(x))*sum(x[1,])
    fe_4 <- (sum(x[,1])/sum(x))*sum(x[2,])
    fe_5 <- (sum(x[,2])/sum(x))*sum(x[2,])
    fe_6 <- (sum(x[,3])/sum(x))*sum(x[2,])

    # Observed Frequencies
    fo_1 <- x[1,1]
    fo_2 <- x[1,2]
    fo_3 <- x[1,3]
    fo_4 <- x[2,1]
    fo_5 <- x[2,2]
    fo_6 <- x[2,3]

    # (Observed - expected)^2
    fo_fe_sq1 <- (fo_1-fe_1)^2
    fo_fe_sq2 <- (fo_2-fe_2)^2
    fo_fe_sq3 <- (fo_3-fe_3)^2
    fo_fe_sq4 <- (fo_4-fe_4)^2
    fo_fe_sq5 <- (fo_5-fe_5)^2
    fo_fe_sq6 <- (fo_6-fe_6)^2

    # ((Observed - expected)^2)/expected
    chi_square1 <- fo_fe_sq1/fe_1
    chi_square2 <- fo_fe_sq2/fe_2
    chi_square3 <- fo_fe_sq3/fe_3
    chi_square4 <- fo_fe_sq4/fe_4
    chi_square5 <- fo_fe_sq5/fe_5
    chi_square6 <- fo_fe_sq6/fe_6

    # Chi-square statistic
    chi_square <- chi_square1 + chi_square2 + chi_square3 + chi_square4 +
      chi_square5 + chi_square6
  }
  print(chi_square)
}
chi_square <- chi_square_by_hand(cont_table)
```

$\chi 2 = 3.79$

(b) Now calculate the p-value from the test statistic you just created (in R).[2] What do you conclude if $\alpha = 0.1$?

```
p_value <- pchisq(chi_square, df=(2-1)*(3-1), lower.tail=FALSE)
p_value
```

The p-value was found to be 0.150. We conclude that there is not sufficient evidence to reject the null hypothesis that the variables class and solicitation of bribe are statistically independent as our p-value is greater than the alpha of 0.1.

(c) Calculate the standardized residuals for each cell and put them in the table below.

```
chi_in_r <- chisq.test(cont_table)
chi_in_r$stdres
```

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.322	-1.642	1.523
Lower class	-0.322	1.642	-1.523

(d) How might the standardized residuals help you interpret the results?

Standardized residuals tell us the number of standard errors the difference in the observed and expected values fall from 0, which we expect to be the difference when our null hypothesis is true. This information gives us another level of evaluation when determining the conclusions of our results. In a large sample size, the probability distribution of each of the residuals follows a normal distribution, meaning that a value that falls outside of approximately +/-2 has a very small likelihood of occurring (small p-value), and therefore can be considered significant in rejecting the independence of the variables. In this circumstance, none of the values are outside of this range, reinforcing our statement earlier that we cannot reject the null hypothesis that the variables of class and solicitation of bribe are statistically independent. These residuals also give us insight on each level of the variable, allowing us to the note variation in the independence of each value in the sample.

[2]Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

Name	Description
`GP`	An identifier for the Gram Panchayat (GP)
`village`	identifier for each village
`reserved`	binary variable indicating whether the GP was reserved for women leaders or not
`female`	binary variable indicating whether the GP had a female leader or not
`irrigation`	variable measuring the number of new or repaired irrigation facilities in the village since the reserve policy started
`water`	variable measuring the number of new or repaired drinking-water facilities in the village since the reserve policy started

[3] Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica.* 72 (5), 1409-1443.

(a) State a null and alternative (two-tailed) hypothesis.

The null hypothesis is that the reservation policy and number of new or repaired drinking water facilities in the villages are not linearly related, meaning that the slope (beta) is equal to zero. The alternative hypothesis is that the reservation policy and number of new or repaired drinking water facilities in the villages are linearly related, meaning that the slope (beta) is not equal to zero.

(b) Run a bivariate regression to test this hypothesis in R (include your code!).

```r
regression <- lm(water ~ reserved, data = df)
summary(regression)
```

```
Call:
lm(formula = water ~ reserved, data = df)

Residuals:
    Min      1Q  Median      3Q     Max 
-23.991 -14.738  -7.865   2.262 316.009 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   14.738      2.286   6.446 4.22e-10 ***
reserved       9.252      3.948   2.344   0.0197 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 33.45 on 320 degrees of freedom
Multiple R-squared:  0.01688,	Adjusted R-squared:  0.0138 
F-statistic: 5.493 on 1 and 320 DF,  p-value: 0.0197
```

The intercept is 14.738 and the slope is 9.252. The regression equation is: Y = 14.738 + 9.252x.

(c) Interpret the coefficient estimate for reservation policy.

The coefficient estimate for reservation policy is 9.252. This value means that for every 1 unit increase in the reserved variable (x), the water variable (y) increases by 9.252. In regards to the null hypothesis, this value suggests, along with the p-value of 0.0197 and large test statistic of 2.344, that at a 95% confidence level we can reject the null hypothesis that the slope (beta) is zero and that the reservation policy and number of new or repaired drinking water facilities in the villarges are not linearly related. In the specific contexts of this study, the coefficient estimate indicates that when the Gram Panchayat (GP) goes from a policy of no reservation for female council heads to a policy reserving seats for female council heads, the number of new or repaired drinking-water facilities in the village increases by an average of 9.252 facilities.